

CSS

Background, h&w , flex, inline-block,

Theory

1. Borders in CSS

Definition: Borders define the outline around an element. **Use case:** Used for styling boxes, headings, and layouts. **Key point:** Can be customized with width, style, and color.

Syntax:

```
h1 {  
  border: 5px blue groove; /* shorthand */  
}
```

2. Padding and Margin

Definition:

- **Padding** → space inside the border, around content.
- **Margin** → space outside the border, between elements. **Key point:** Padding affects element size; margin affects spacing between elements.

Syntax:

```
h1 {  
  padding: 10px 20px;  
  margin: 20px 0;  
}
```

3. Display Property

Definition: Controls how elements are rendered. **Types:**

- **block** → takes full width.
- **inline** → flows with text, ignores width/height.
- **inline-block** → inline but respects width/height.

Syntax:

```
p { display: inline-block; }
```

4. Flexbox

Definition: A layout model for arranging elements in rows or columns. **Use case:** Responsive design, alignment, spacing. **Key point:** Controlled with justify-content, align-items, flex-direction.

Syntax:

```
section {  
  display: flex;  
  flex-direction: column;  
  justify-content: center;  
  align-items: center;  
}
```

5. Box Shadow & Border Radius**Definition:**

- box-shadow → adds shadow around elements.
- border-radius → rounds corners.

Syntax:

```
h1 {  
  box-shadow: 0 0 5px #333;  
  border-radius: 10px;  
}
```

Coding Round Questions

- Border styles → Demonstrate solid, dashed, dotted, and groove borders.
- Padding vs Margin → Show difference between padding and margin with two boxes.
- Display property → Create examples of block, inline, and inline-block elements.
- Flexbox layout → Arrange three boxes horizontally and center them using flexbox.
- Box shadow → Add a shadow to a heading element.

Challenge

Problem: Product Card Layout

Requirements:

- Create a product card with:
 - Image, title, description, and price.
- Style with:
 - Border, padding, margin.
 - Flexbox for alignment.
 - Box shadow and border radius for aesthetics.
- Ensure responsiveness with flexbox.



